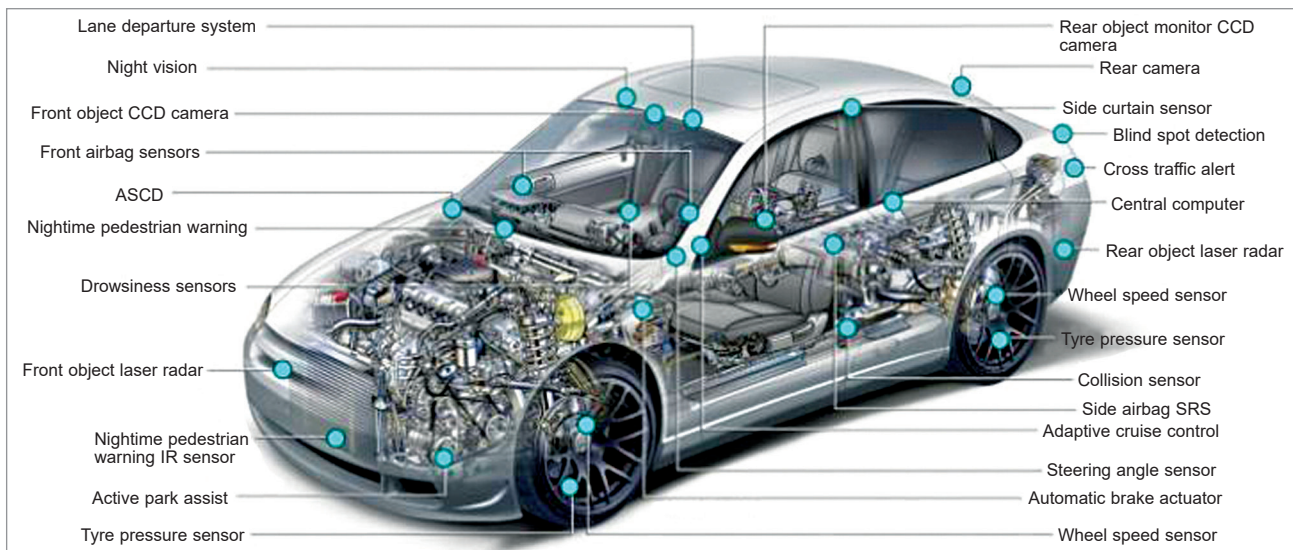




Vehicle sensors

features, and the overall ticket size of electronic components is increasing. For example, in an A-segment car, a wireless charger may or may not be present, but in a C- or D-segment, it will be standard. We have not even come close to the ceiling of electronic components being equipped in vehicles. "This will only escalate from here," he states.

Among the list of components Pioneer is looking to procure locally from India, camera sensors, systems on chips (SoCs), memories, and displays rank at the top. Until now, the company has been selling products made outside India in the Indian market.

Dr Siva Subramanian, Chief Innovation Officer at Pioneer Corporation, mentions in a conversation with EFY, "Our plan is to start manufacturing products for the Indian and global markets in India right away. We have already begun collaborating with Indian design and manufacturing partners. Given the growth potential, there is a strong possibility that we will establish a manufacturing facility in India in the future."

A report by Mordor Intelligence highlights that the 'India Passenger Car Market' size will be an estimated \$39.82 billion in 2024 and is expected

to reach \$53.04 billion by 2029, growing at a CAGR of 5.90% during the forecast period (2024-2029).

Focus on the child parts

Whether it is PCBs, heat exchangers, thin-wall casings, or thermoplastics, manufacturers of components usually have to depend on imports to meet their needs. In the true sense, thin-wall casing used in electric control units (ECUs) accounts for 20% to 25% of this component's value. Additionally, heat exchangers contribute a 20% share in a thermal management system.

If ACMA is to be believed, then making these two parts locally in India can reduce the country's import bill by \$275 million by 2032. Also referred to as child parts, these find usage not only in the automotive industry but in other industries as well, including drones, robotics, consumer electronics, and more! Focusing on child parts makes more sense because the automotive industry's evolution from 'parts per million to parts per billion' is kept in consideration.

Bohra explains, "There was a time when there was a tolerance for quality issues whenever you were buying from a supplier. The same

would be noticed on an assembly line. Today, that tolerance is referred to as parts per billion instead of parts per million. For example, earlier, one part per million was supposed to be okay if there was an issue. Today, even that is not expected."

According to him, export is also a significant opportunity for India. At the same time, domestic demand is skyrocketing in India. "It will take some time for the Indian industry to grow its international share of revenue because it cannot compromise on the delivery targets set by the domestic players," he said.

Manek suggests that software is not just a layer added to electronics used in cars but also an opportunity for India's software industry. Notably, a majority of automotive electronics products with medium to high software requirements are currently imported.

He says, "From a local vendor base, it is a significant opportunity. Many OEMs are vouching for these opportunities. Now the question is how do we capitalise on these opportunities for everyone's benefit." **EFY**

Mukul Yudhveer Singh is the Editor for electronicsb2b.com